Evolution-Trained Feed-Forward Neural Control for UAV Target Navigation in Multi-Agent Swarms

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1 Abstract

Autonomous path planning and collision avoidance are critical capabilities for modern Unmanned Aerial Vehicle (UAV) fleet management. This paper presents a simulation framework developed in Modelica to model multi-UAV dynamic environments and evaluate learning-based flight control policies. A feed-forward neural network (FFNN) is trained using OpenAI Evolution Strategies (ES) to navigate unknown spaces toward target coordinates while avoiding collisions with neighboring UAVs. Using a simplified point-mass model integrated with a simulated sparse LiDAR perception grid, the proposed control loop demonstrates effective decentralized local navigation across a N-UAV swarm architecture.

2 Introduction

Unmanned Aerial Vehicles (UAVs) have rapidly evolved from single-operator remotely piloted platforms into fully autonomous multi-vehicle fleets capable of executing complex missions in logistics, search and rescue, dynamic environmental monitoring, and perimeter defense. As fleet density increases and missions require operation in cluttered, unmapped environments, dynamic local navigation—specifically autonomous path planning and high-fidelity collision avoidance—becomes a fundamental prerequisite for safe flight. Safe operation demands real-time decision-making capabilities that allow individual vehicles to adapt to moving obstacles and non-cooperative neighbor traffic without human intervention.

Research has heavily shifted toward decentralized control paradigms, where each UAV computes its own motion commands using onboard sensory information and local neighbor states. Decentralized navigation significantly enhances scalability and communication bandwidth efficiency, but it introduces non-trivial local-minima hazards and unpredictability when multi-agent collision avoidance behaviors interact.

This paper presents a simulation framework developed in Modelica to model multi-UAV dynamic environments and evaluate learning-based flight control policies. A feed-forward neural network (FFNN) is trained directly against Modelica simulations using OpenAI Evolution Strategies (ES) to navigate unknown environments toward target coordinates. Although the overall framework supports decentralized multi-agent dynamics, our core focus is evaluating a single target UAV governed by the trained FFNN as its mission is to successfully reach a designated target while navigating through an active swarm environment where the remaining $N-1$ UAVs fly in continuously changing, random trajectory patterns. Integrated with a simulated sparse LiDAR perception grid and a simplified point-mass dynamic model, the proposed control loop demonstrates effective real-time obstacle and multi-UAV collision avoidance across a N-UAV swarm architecture.

3 Literature Review

The model and training methodology are grounded in established literature on cyber-physical simulation and reinforcement learning for autonomous navigation. Modelica was selected for its strength in acausal modeling of physical system dynamics governed by differential algebraic equations (DAEs). The neural network training and optimization adopts the OpenAI Evolution Strategies (ES) framework [1] a scalable, gradient-free black-box optimization algorithm suitable for environments with non-differentiable simulation boundaries or external C-function bindings.

4 System Methodology & Simulation Architecture

To balance physical fidelity with computational performance, each UAV is modeled as a 3-DOF point-mass dynamic model rather than a fully articulated multi-rotor body. Kinetic and kinematic constraints are preserved to ensure realistic velocity and acceleration limits. The baseline architecture scales to an N=5 UAV swarm, managed by a centralized SWARM supervisor block. Each individual UAV consists of three internal modules:

- **SETPOINT (AutoPilot Target Generator)**: Provides high-level goal vectors as target spherical velocity coordinates to the controller.

- **CTR (Flight Control Module)**: Functions as the primary onboard processing unit. It converts incoming spherical velocities into Cartesian coordinates, computes neural network inference outputs every 0.02 seconds, and generates target thrust commands.
- **UAV-HARDWARE (Physical Dynamics)**:Accepts calculated thrust values to update the physical state vectors (Cartesian position, velocity, and acceleration) via dynamic equations of motion. The maximum velocity value is set to 3 m/s

4.1 UAV System

The parameters set for modeling each UAV are based on realistic physical and commercial specifications. These constants govern the aerodynamic drag, total mass, and electrical storage capacity of the aircraft during simulation:

- **Mass & Aerodynamics:**
  - **Mass (m):** 5.0 kg — Total mass of the UAV platform.
  - **Drag Coefficient (C_d):** 0.5 — Dimensionless aerodynamic drag coefficient.
  - **Frontal Area (A_{side}):** 0.15 m^2 — Frontal reference area subjected to lateral airflow.
  - **Top Area (A_{top}):** 0.30 m^2 — Top reference area subjected to vertical airflow.
  - **Air Density (ρ):** 1.225 kg/m^3 — Standard sea-level air density.
- **Electrical & Power System:**
  - **Nominal Voltage (V_{nom}):** 22.2 V — Nominal operating voltage for a 6S LiPo battery pack.
  - **Battery Capacity (C_{bat}):** 16.0 Ah — Total energy storage capacity.
  - **Stored Energy (E_{Wh} / E_{J}):** 355.2 Wh (1.27872×10^6 J) — Total nominal energy available.
  - **Hover Power (P_{hover}):** 750.0 W — Estimated electrical power drain during static hover.

4.2 Perception & Sensing

Local perception is simulated via an onboard LiDAR system modeled in the FRONTIER block. The perceptual field follows a Vector Field Histogram (VFH) [2] sparse grid structure:

- Detection Range: 20 m
- Horizontal Field of View (FoV): 180°
- Vertical Field of View (FoV): 60°
- Grid Resolution: 3 rows × 9 columns: 27 cells

Spatial vectors are expressed in the UAV's local frame of reference using an onboard compass for target heading estimation. The threshold used for establishing the contact with the target is 0.5 meters

4.3 Controller Architecture (FFNN Layer Integration)

The neural policy is implemented as a Feed-Forward Neural Network (FFNN) written in standard C and compiled into Modelica using external function calls invoked every T = 0.02s.
Parameters Breakdown:

- **Input Layer (33 nodes)**:
  - 27 inputs: perception state from the LiDAR grid.
  - 3 inputs: relative target Cartesian displacement $(\Delta x, \Delta y, \Delta z)$.
  - 3 inputs: current cartesian velocity vector components $(v_{x,\text{cmd}}, v_{y,\text{cmd}}, v_{z,\text{cmd}})$.
- **Hidden Layer (16 nodes)**: Fully connected with 16 bias parameters.
- **Output Layer (3 nodes)**: Commanded Cartesian velocity vector components $(v_{x,\text{cmd}}, v_{y,\text{cmd}}, v_{z,\text{cmd}})$.

Total Parameters : $(33 \times 16 + 16) + (16 \times 3 + 3) = 595$

5 Optimization & Training Procedure

Parameter optimization of the 595-weight neural controller is executed via OpenAI Evolution Strategies (ES), a gradient-free black-box optimization framework. To prevent premature convergence and guide policy exploration through complex multi-agent interactions, a three-stage curriculum learning pipeline is implemented across 600 total training epochs (200 epochs per stage). The size of the swarm during training was set as 5, having 1 AI and 4 NON-AI.

5.1 Curriculum Learning Stages

- **Stage 1 (Basic Flight & Goal Acquisition)**:Non-AI UAVs are spawned at a distant offset ($d > 100$ m) to isolate single- AI agent flight mechanics. The fitness function is reduced to proximity bonus (R_{prox}) and hover penalties (P_{hover}), allowing the policy to master stable dynamic flight toward target coordinates B.
- **Stage 2 (Kinetic & Energy Optimization)**: Obstacle UAVs remain distant while energy (P_{energy}) and

time penalties (P_{time}) are activated. Spawn boundaries and target offsets are expanded from 5 m to 25 m. This stage trains the network to enforce speed bounds ($v \leq 3.0$ m/s) and minimize total battery discharge ($\Delta SoC_{\%}$).

- **Stage 3 (Dynamic Swarm & Local Avoidance)**: Non-hero UAVs are dynamically spawned directly along the AI UAV's target corridor. The proximity warning penalty (P_{warning}) is activated and conditioned on active LiDAR detection grid outputs. The policy learns to parse sparse 27-cell visual inputs and execute local evasion maneuvers while preserving goal-seeking trajectory efficiency.

5.2 Antithetic Search & Optimization Loop

Each generation evaluates $N_{\text{pop}} = 32$ perturbed weight variants ($\boldsymbol{\theta}_i = \boldsymbol{\theta} + \sigma\boldsymbol{\epsilon}_i$) generated via antithetic sampling ($\boldsymbol{\epsilon}_{1..16}, -\boldsymbol{\epsilon}_{1..16}$). Rollouts are evaluated across 8 parallel worker threads running compiled C/Modelica executables ('MultiUAVSystem'). The global weight vector $\boldsymbol{\theta}$ is updated using rank-normalized fitness scores A_i and the Adam optimizer:

$$\nabla_{\boldsymbol{\theta}} J \approx \frac{1}{N_{\text{pop}}\sigma} \sum_{i=1}^{N_{\text{pop}}} A_i \boldsymbol{\epsilon}_i$$

Where the values set for finetuning the model were:

- $\alpha = 0.015, \quad \sigma = 0.020$
- $\beta_1 = 0.9, \quad \beta_2 = 0.99$

5.3 Reward Function Formulation

The total reward R for an individual trajectory evaluation is computed as a weighted summation of state bonuses and penalties:

$$R = R_{\text{prox}} - P_{\text{dist}} - P_{\text{overspeed}} - P_{\text{hover}} - P_{\text{energy}} - P_{\text{time}} - P_{\text{warning}}$$

Where:

- $R_{\textbf{prox}}$: Bonus for closing the distance to the assigned goal coordinate.
- $P_{\textbf{dist}}$: Penalty proportional to remaining distance to target at rollout completion.
- $P_{\textbf{overspeed}}$: Penalty for exceeding structural kinetic velocity thresholds.
- $P_{\textbf{hover}}$: Penalty penalizing stationary/zero-velocity hovering behaviors.
- $P_{\textbf{energy}}$: Penalty penalizing excessive thrust fluctuations.
- $P_{\textbf{time}}$: Constant step penalty encouraging time-optimal navigation.
- $P_{\textbf{warning}}$: Proximity penalty enforced when encroaching upon obstacle or neighbour buffer

6 Preliminary Findings & Discussion

The **first** [1] batch of tests was conducted on 5 different scenarios, each one compressing or stretching the scope of the environment.

1. **Dense Corridor Low Alt**
2. **High Altitude Ascent**
3. **Fast Cross Traffic**
4. **Extreme Distance**
5. **Really Extreme Distance**

The number of UAVs was kept as per training settings. The results demonstrate the remarkable effectiveness of the neural control system, highlighting advanced navigation, obstacle avoidance, and stabilization capabilities.

- **Collision Avoidance and Success Rates**: The parametric tests recorded an extremely low percentage of collisions between the piloted UAV and the rest of the swarm, which featured a spawn configuration based on a circular pattern. Analysis of the summary table shows that across five stress test scenarios, impacts with obstacles were minimal: only 2 collisions in the **Dense Corridor Low Alt** and 1 collision in both **High Altitude Ascent** and **Extreme Distance**. In the **Fast Cross Traffic** scenario, the drone recorded an absolute zero in terms of collisions, achieving 100 % success rate. In none of the scenarios did a ground crash occur (**Ground Crashes**).
- **Generalization and Extrapolation Capabilities**: A fundamental finding from the simulation is the ability of the Feed-Forward Neural Network (FFNN) to generalize the flight environment. Although training was limited to a confined operational space, defined by the condition current_max_dist < 25.0, the agent successfully piloted the UAV in test scenarios with significantly greater distances. This shows that the model learned the relational dynamics between the target and obstacles rather than simply memorizing the coordinates of the training ground.
- **Approach Dynamics and Hovering**: Unlike classic "greedy" logic, where the agent tends to dash at maximum speed toward the target, the UAV adopts a balanced cruising strategy. The FFNN guides the drone smoothly through the swarm and then stabilizes it with

precision. This fine-tuning is reflected in the high **Stable Hover** rates, ranging from 80.0 %to 100.0 % in scenarios where the target is reached. The agent does not frantically overshoot the objective, but maintains hovering with average drifts (**Avg Hover Drift**) contained between 0.59 m and 0.83 m.

- **Energy Efficiency and Balanced Gait**: The conservative and fluid behavior of the drone is the direct result of the fitness function architecture used during the training epochs. The specific weighting of the penalties forced the agent to develop a balanced gait, avoiding excessive speeds in the airspace and thus preserving the system's energy capacities.

- **Physical and Actuation Limits**: The observed low energy output is intrinsically supported and constrained by the Modelica model architecture. Thrust commands cannot grow indefinitely: in the **CTR** block, the calculated forces undergo a limitation based on the maximum allowed tilt and are saturated at the maximum Thrust value. Furthermore, the output signals from the last layer of the neural network are scaled using a hyperbolic tangent function (**tanh**), which mathematically limits the local Cartesian velocity setpoints to a safe range between -3.0 and +3.0 m/s. Finally, the UAV hardware integrates realistic differential equations, where the thrust (for example, on the Z axis) must constantly counteract gravity and aerodynamic drag.

The **second** [2] batch of tests was focused on 3 specific parameter variations:

1. **Number of non-AI UAVs**: 3 to 5 to 9
2. **Distance Spawn**: 4 different settings
   - **Narrow**: ranging from 5 to 9 meters
   - **Close**: ranging from 10 to 15 meters
   - **Medium**: ranging from 30 to 45 meters
   - **Extreme**: ranging from 70 to 100 meters
3. **Simulation Time**: 60 to 120 to **Unlimited** seconds

50 simulation runs were performed per total parameter combinations, amounting to a **total** of 450 runs.

7 Conclusion

The framework proved to be highly performative in simulating multi-UAV dynamic environments and training a generalized Feed-Forward Neural Network to complete autonomous path planning. The results successfully demonstrate that the trained FFNN correctly maneuvers the single UAV in critical target-reaching missions, effectively avoiding dynamic traffic. Furthermore, this study highlights the strong compatibility of coupling Modelica as a rigorous simulation environment with Evolution Strategies (ES) for training robust aerospace control systems. The next step extends this framework to the non-AI UAVs, enabling them to operate as a pursuit swarm tasked with intercepting the AI-guided UAV before it reaches its target.

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Table 1: **Simulation Scenarios and Initialization Parameters**

Scenario	**Hero Z Range** [m]	**Target Dist.** [m]	**Target ΔZ** [m]	**Obs. Spawn Radius** [m]	**Obs. Z Range** [m]	**Obs. V_r Range** [m/s]
1. Dense Corridor Low Alt	$[2.0, 5.0]$	$[8.0, 15.0]$	$[-1.0, 2.0]$	$[2.0, 8.0]$	$[1.5, 6.0]$	$[1.5, 3.0]$
2. High Altitude Ascent	$[1.0, 3.0]$	$[15.0, 30.0]$	$[15.0, 25.0]$	$[5.0, 15.0]$	$[5.0, 20.0]$	$[1.0, 2.5]$
3. Fast Cross Traffic	$[5.0, 10.0]$	$[12.0, 22.0]$	$[-3.0, 3.0]$	$[4.0, 12.0]$	$[4.0, 12.0]$	$[2.5, 4.0]$
4. Extreme Distance 3D	$[12.0, 30.0]$	$[25.0, 45.0]$	$[-5.0, 15.0]$	$[8.0, 25.0]$	$[2.0, 25.0]$	$[0.8, 2.2]$
5. Really Extreme Dist. 3D	$[40.0, 60.0]$	$[70.0, 100.0]$	$[-5.0, 15.0]$	$[23.0, 70.0]$	$[12.0, 60.0]$	$[2.5, 3.0]$

Table 2: **Simulation Results for Different Swarm Sizes** ($N = 3, 5, 10$)

Sim Time [s]	**Spawn Tier**	**Success Rate [%]**	**Avg Target Time [s]**
Total Drones: 3 (2 Active Obstacles)			
60.0	Close	98.00	16.98
120.0	Close	98.00	16.98
Unlimited	Close	98.00	16.98
60.0	Medium	100.00	38.20
120.0	Medium	100.00	38.20
Unlimited	Medium	100.00	38.20
60.0	Extreme	2.00	56.62
120.0	Extreme	100.00	70.52
Unlimited	Extreme	100.00	70.52
Total Drones: 5 (4 Active Obstacles)			
60.0	Close	92.00	16.61
120.0	Close	92.00	16.61
Unlimited	Close	92.00	16.61
60.0	Medium	100.00	37.57
120.0	Medium	100.00	37.57
Unlimited	Medium	100.00	37.57
60.0	Extreme	4.00	58.61
120.0	Extreme	96.00	69.35
Unlimited	Extreme	96.00	69.35
Total Drones: 10 (9 Active Obstacles)			
60.0	Close	86.00	15.98
120.0	Close	86.00	15.98
Unlimited	Close	86.00	15.98
60.0	Medium	100.00	37.91
120.0	Medium	100.00	37.91
Unlimited	Medium	100.00	37.91
60.0	Extreme	4.00	58.62
120.0	Extreme	100.00	70.68
Unlimited	Extreme	100.00	70.68

References

[1] Tim Salimans et al. *Evolution Strategies as a Scalable Alternative to Reinforcement Learning*. 2017. arXiv: 1703.03864 [stat.ML]. URL: https://arxiv.org/abs/1703.03864.

[2] J. Borenstein and Y. Koren. "The vector field histogram-fast obstacle avoidance for mobile robots". In: *IEEE Transactions on Robotics and Automation* 7.3 (1991), pp. 278–288. DOI: 10.1109/70.88137.